

Total No. of Questions : 8]

SEAT No. :

PE-2583

[Total No. of Pages : 2

[6583]-114

T.E. (Electronics & Computer Engineering)

DATA COMMUNICATION

(2019 Pattern) (Semester - V)(310343)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Neat diagrams must be drawn wherever necessary.

- Q1)** a) Explain ASK Transmitter and Receiver with waveform. [5]
b) Explain QPSK Transmitter in detail with the help of block diagram. [6]
c) Draw and explain BPSK modulator and demodulator with its waveforms. [6]

OR

- Q2)** a) Explain QPSK Receiver in detail with the help of block diagram. [5]
b) Explain M-ary PSK in detail with the help of Transmitter and Receiver. [6]
c) Compare between FSK and PSK. [6]

- Q3)** a) Compare between CDMA and FDMA. [5]
b) Explain CDMA technology in details. [6]
c) Explain FDMA in details. Also State its Advantages & Disadvantages. [6]

OR

- Q4)** a) What is PN Sequence in CDMA? Explain [5]
b) Explain TDMA technology in details. Also state its advantages and disadvantages. [6]
c) What are the Advantages & Disadvantages of Code Division Multiple Access? [6]

P.T.O.

- Q5)** a) Prove the following statement, If receiver know the message being transmitted, the amount of information carried is "Zero" [5]
- b) Explain the following terms, [6]
- Measure of Information
 - Unit of Information
 - Information Rate
- c) In binary sequence '0' occur with probability of $\frac{1}{2}$ and '1' occur with probability of $\frac{2}{3}$ then calculate amount of information conveyed by each binit. [7]

OR

- Q6)** a) Explain Huffman coding algorithm. Also state its advantages. [5]
- b) Let us consider a binary source with source alphabet $s = \{S_1, S_2\}$ with probabilities $P = \{1/256, 255/256\}$ find entropy. [6]
- c) Define entropy. Drive the expression for entropy. [7]
- Q7)** a) Define and explain single bit error and burst error. Also state disadvantages of error control coding. [5]
- b) Define the following related to error control coding. [6]
- code word
 - code rate
 - code efficiency
 - hamming distance
 - Minimum distance
- c) What is meant by error detection? Explain check sum error detection method. [7]

OR

- Q8)** a) Define error control coding. Explain need of error control coding. [5]
- b) What is meant by error detection? Explain parity checking error detection method. [6]
- c) Find the checksum of following message 10001001, 11100010, 001001001, 10000100. [7]

